



S H E N T O N
C O L L E G E

ATAR CHEMISTRY

UNITS 1 & 2

TEST 1

**Properties and Structure of Atoms
and
Bonding**

MULTIPLE CHOICE SECTION

DO NOT WRITE ON THIS BOOKLET

PLEASE DO NOT TURN THE PAGE UNTIL INSTRUCTED TO DO SO

Total Recommended time: 50 minutes

PART A: MULTIPLE CHOICE SECTION

Place a cross (X) through your answer on your answer booklet. DO NOT write in this booklet.

1. A sulfate ion (SO_4^{2-}) contains the isotopes S-33 and O-15. How many electrons, protons and neutrons does the ion possess?

	electrons	protons	neutrons
a.	46	48	48
b.	93	91	96
c.	50	48	45
d.	48	45	50

2. An unknown solid material is added to water and stirred. On filtration a residue remains on the filter paper. Further experiments show that the quantity of residue is unaffected by addition of more water. When the filtrate (the solution that went through the filter paper) is evaporated, a solid residue is found in the container. From this information the original unknown material must have been:

- a. a mixture
- b. an element
- c. a pure substance
- d. a compound

3. Which of the following formulae represents a molecule of a compound?

- a. S_8
- b. NO_2
- c. MgCl_2
- d. Si

4. The ion X^{3+} has an electronic configuration 2,8. Therefore element X is in:

- a. Group 2, Period 3
- b. Group 13, Period 2
- c. Group 13, Period 3
- d. Group 14, Period 2

5. Which of the following would conduct an electric current?

- I a crystal of sodium chloride
- II molten silver nitrate
- III an aqueous solution of ammonium sulfate
- IV magnesium

- a. I, II and III only
- b. II, III and IV only
- c. II and IV only
- d. II and III only

For questions 6 – 8 use the following representation of a Periodic Table.

I	II											III	IV	V	VI	VII	VIII
																	K
J	A											L	F	M		E	
										B	D						
																H	G

Elements are NOT listed by their usual symbols.

6. Which of the following pairs of elements is least likely to form compounds?
 - a. A and E
 - b. M and F
 - c. B and D
 - d. G and K

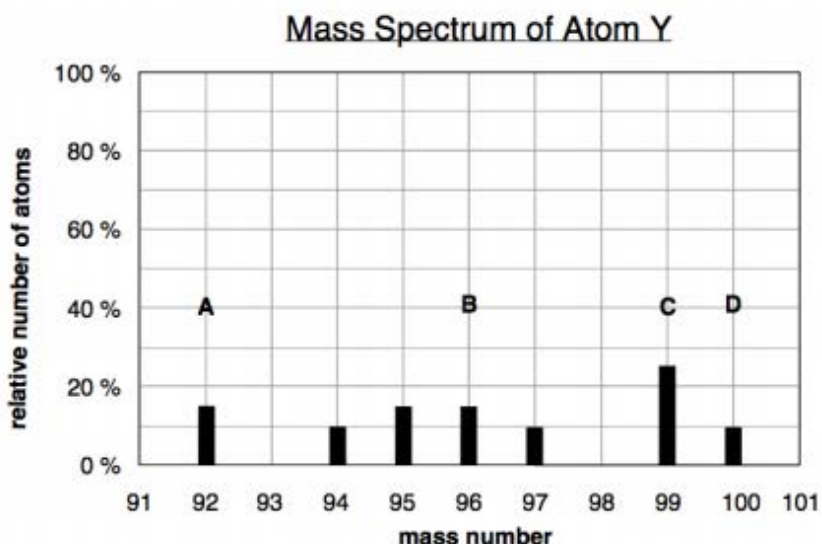
7. Which of the following elements could have a melting temperature of 660°C and be a good conductor of electricity as a solid?
 - a. E
 - b. L
 - c. G
 - d. F

8. Which of the following pairs of elements would form a covalent molecular compound?
 - a. J and A
 - b. D and L
 - c. E and K
 - d. M and E

9. Which of the following is INCORRECT?
 - a. Mixtures cannot be separated by physical means.
 - b. Elements and compounds are pure substances.
 - c. Sublimation refers to the phase change of a solid to a gas.
 - d. Mixtures can be homogeneous or heterogeneous.

10. Listed below are the atomic numbers of different elements. Which of the sets of elements contains atoms with similar properties?
 - a. 1, 2 and 3
 - b. 1, 3 and 11
 - c. 7, 16 and 35
 - d. 13, 31 and 49

11. Based on the mass spectrum of atom Y below, which of the following statements is FALSE?



- a. Peak A and Peak D come from atoms that have the same number of electrons
- b. There are 7 isotopes of atom Y
- c. Peak C comes from the most abundant isotope of atom Y
- d. Peak D comes from an atom with 4 more protons than the atom that gave Peak B

12. Which of the following lists the steps in mass spectrometry in the correct order?

- a. Detection, ionisation, acceleration, deflection
- b. Ionisation, deflection, acceleration, detection
- c. Ionisation, acceleration, deflection, detection
- d. Acceleration, ionisation, deflection, detection

13. Which of the following statements regarding nanotechnology is TRUE?

- a. Nanomaterials have different physical properties to the bulk material
- b. Objects on a nanoscale can still be seen with the naked eye
- c. Nanomaterials are materials that have particles between 100 and 1000 nm in size
- d. Nanotechnology utilises the fact that atoms in the centre of a material are more reactive than atoms on the surface of a material.

14. Which of the following mixtures could best be separated using simple (not fractional) distillation to obtain both components?

- a. Crude oil and diesel
- b. Sand and water
- c. Sugar and sand
- d. Salt solution

15. Copper wire may be bent easily without breaking. The best explanation of this is that:

- a. Slight changes in relative positions of adjacent copper particles do not break the metallic bonds as these are equally strong in all directions.
- b. The forces between copper atoms are weak, allowing copper particles to be easily moved around.
- c. Copper particles are strongly bonded in layers but only weakly bonded between layers.
- d. Copper particles are arranged in flat molecules that freely slide over each other, allowing material to be bent.

END OF SECTION A